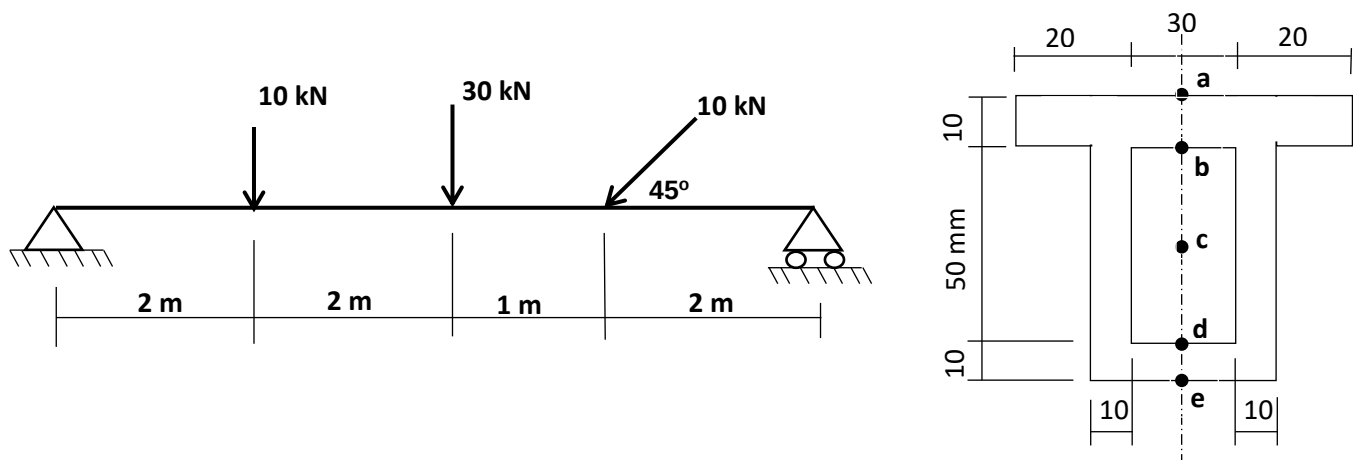
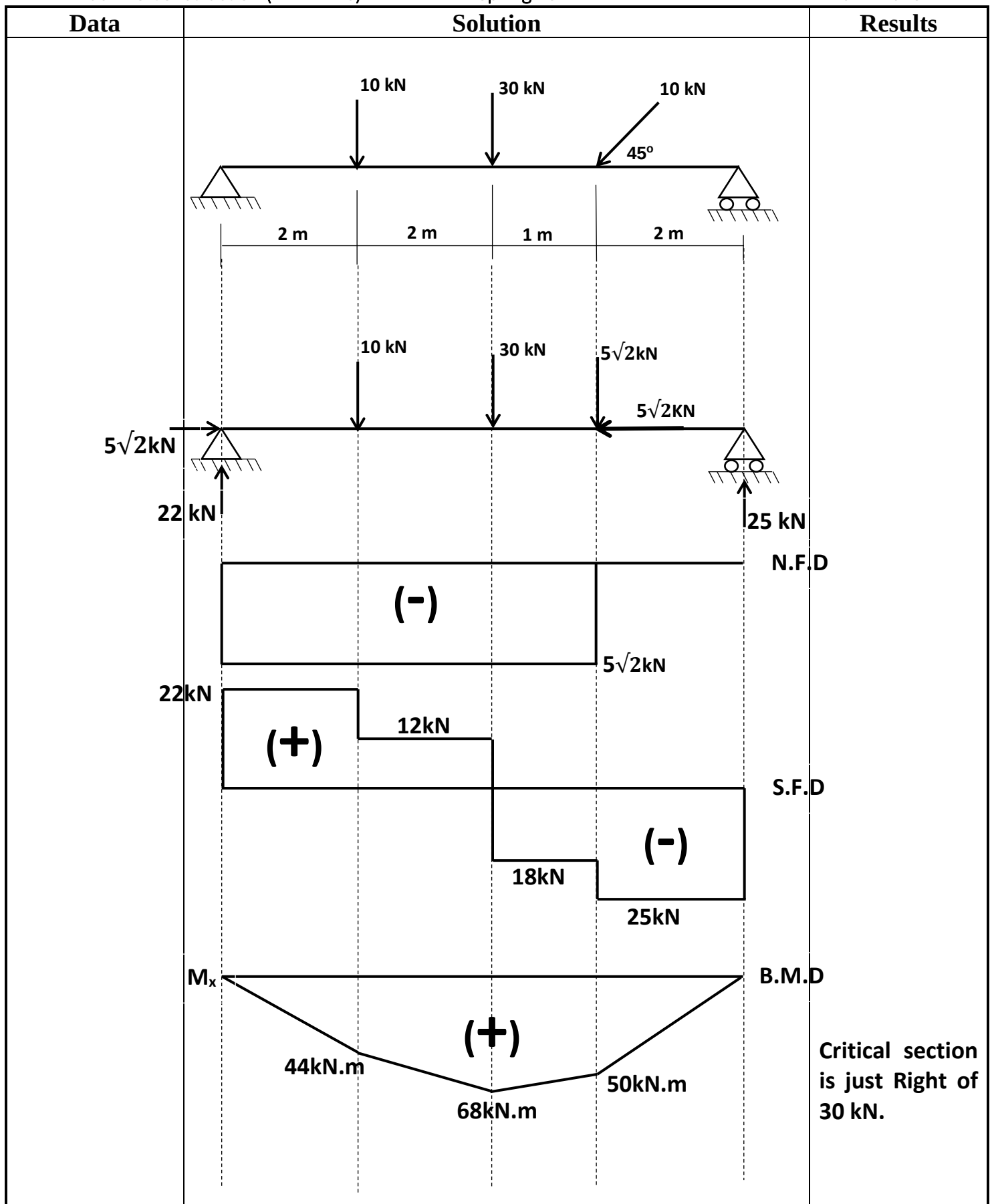
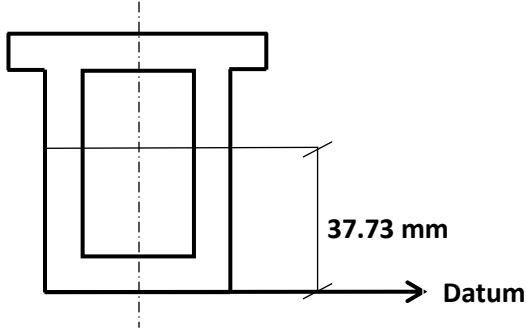


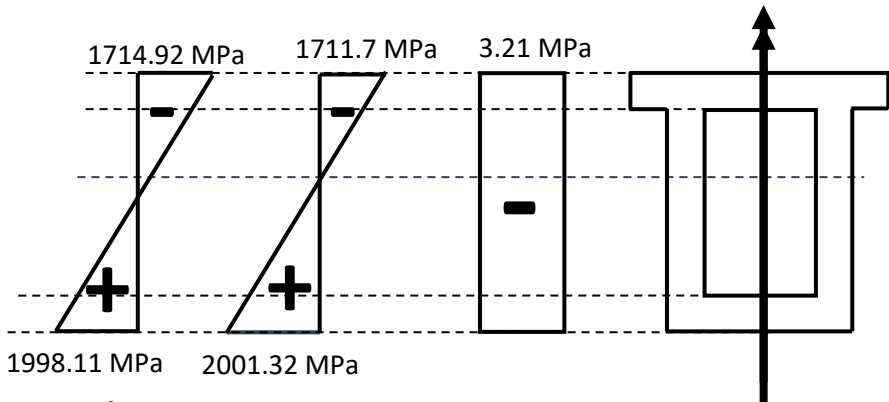
Solved Example

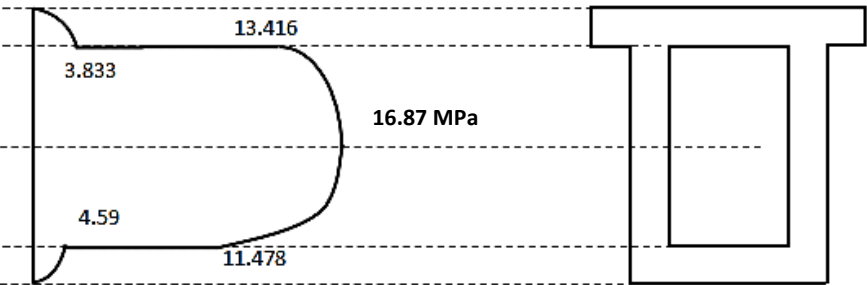
1. **Draw** the normal force, the shearing force, and the bending moment diagrams for the following structure.
2. **Calculate** and **draw** the normal and shear stress diagrams at the critical section for the given cross section.
3. **Draw** the state of stress on an infinitesimal element for the shown points a, b, c, d and e.

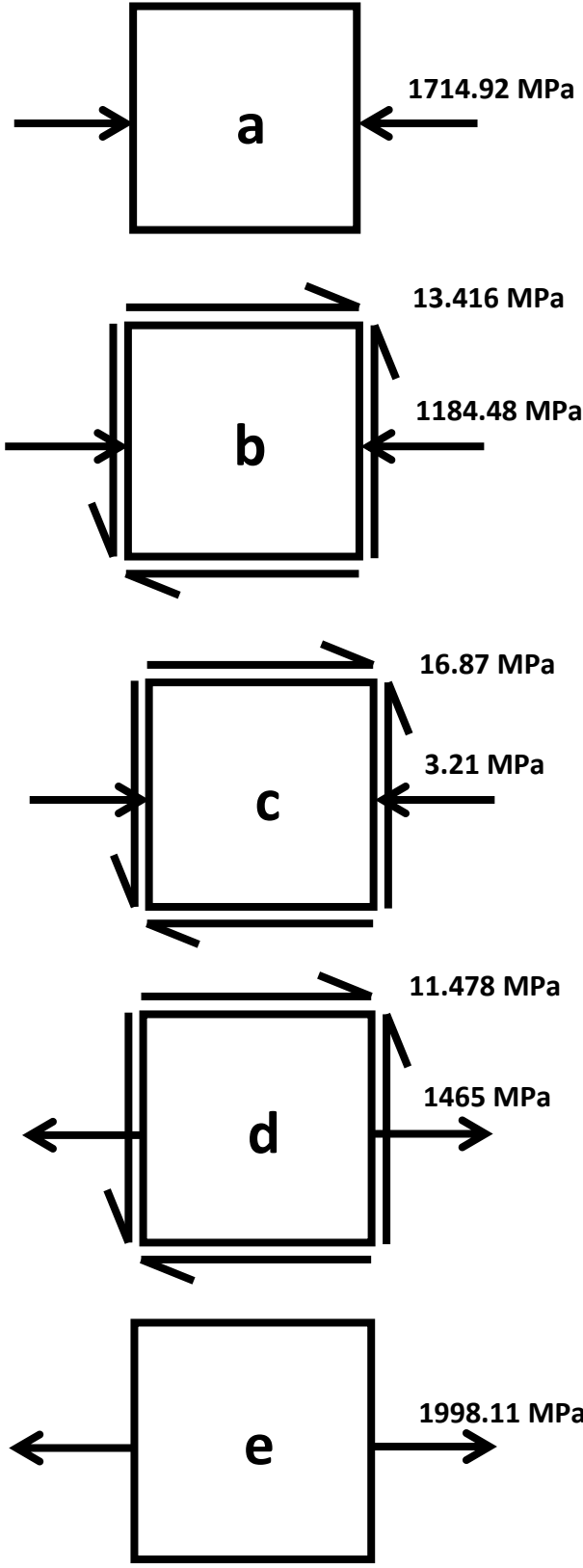




Data	Solution	Results
<u>Loading</u> <u>Conditions:</u> $F = -5\sqrt{2} \text{ kN}$ $V_x = \text{Zero}$ $V_y = 1.8 \text{ kN}$ $M_x = 68 \text{ kN.m} \uparrow$ $M_y = \text{zero}$ $T = \text{zero}$	<u>Area properties:</u> 1. Position of centroid: $\bar{y} = \frac{\sum A_i y_i}{\sum A_i}$ $\bar{y} = \frac{(10)(70)(65) + (50)(60)(30) - (30)(50)(35)}{(10)(70) + (50)(60) - (30)(50)}$  $\sum A_i = (10)(70) + (50)(60) - (30)(50) = 2200 \text{ mm}^2$ 2. Moment of Inertia: $I_x = \frac{(70)(10)^3}{12} + (10)(70)(65 - 37.73)^2 + \frac{(50)(60)^3}{12}$ $+ (50)(60)(37.7 - 30)^2 - \frac{(30)(50)^3}{12} - (30)(50)(37.7 - 35)^2$	$\bar{y} = 37.73 \text{ mm}$ $A = 2200 \text{ mm}^2$ $I_x = 1281971.33 \text{ mm}^4$

Data	Solution	Results
	<u>Stress Calculations:</u> $\sigma = \pm \frac{F}{A} \pm \frac{M_x y}{I_x} \pm \frac{M_y x}{I_y}$  Normal Stress $\sigma_a = -\frac{5\sqrt{2} \times 10^3}{2200} - \frac{68 \times 10^6 \times (70 - 37.73)}{1281971.33}$ $\sigma_b = -\frac{5\sqrt{2} \times 10^3}{2200} - \frac{68 \times 10^6 \times (60 - 37.73)}{1281971.33}$ $\sigma_c = -\frac{5\sqrt{2} \times 10^3}{2200}$ $\sigma_d = -\frac{5\sqrt{2} \times 10^3}{2200} + \frac{68 \times 10^6 \times (37.7 - 10)}{1281971.33}$ $\sigma_e = -\frac{5\sqrt{2} \times 10^3}{2200} + \frac{68 \times 10^6 \times (37.73)}{1281971.33}$	$\sigma_a = -1714.92 \text{ MPa}$ $\sigma_b = -1184.48 \text{ MPa}$ $\sigma_c = -3.21 \text{ MPa}$ $\sigma_d = 1465 \text{ MPa}$ $\sigma_e = 1998.11 \text{ MPa}$

Data	Solution	Results
Shear Stress: $\tau = \frac{V_y Q}{I_x t}$	$Q_b = A \times L = (10 \times 70)(65 - 37.7) = 19.11 \times 10^3$ $\tau_{b_a} = \frac{18 \times 10^3 \times 19.11 \times 10^3}{1.28197 \times 10^6 \times 70}$ $\tau_{b_b} = \frac{18 \times 10^3 \times 19.11 \times 10^3}{1.28197 \times 10^6 \times 20}$ $\tau_c = \left[\frac{(18 \times 10^3)}{1.28197 \times 10^6 \times 20} \right] \times [(10 \times 70)(65 - 37.7) + (50)(60 - 37.7)(11.15) - (30)(60 - 37.7)(11.15)]$ $\tau_{d_a} = \frac{18 \times 10^3 \times (10)(50)(37.5 - 5)}{1.28197 \times 10^6 \times 20}$ $\tau_{d_b} = \frac{18 \times 10^3 \times (10)(50)(37.5 - 5)}{1.28197 \times 10^6 \times 50}$ 	$\tau_a = \tau_e = \text{zero}$ $\tau_{b_a} = 3.833 \text{ MPa}$ $\tau_{b_b} = 13.416 \text{ MPa}$ $\tau_c = 16.87 \text{ MPa}$ $\tau_{d_a} = 11.478 \text{ MPa}$ $\tau_{d_b} = 4.59 \text{ MPa}$

Data	Solution	Results
	<u>Elements:</u>  The diagram illustrates five elements, labeled a through e, each represented by a square. Element 'a' has a horizontal force arrow pointing right and a stress value of 1714.92 MPa. Element 'b' has a horizontal force arrow pointing right, a vertical force arrow pointing down on the right side, and a stress value of 1184.48 MPa. Element 'c' has a horizontal force arrow pointing right, a vertical force arrow pointing down on the right side, and a stress value of 3.21 MPa. Element 'd' has a horizontal force arrow pointing right, a vertical force arrow pointing down on the right side, and a stress value of 1465 MPa. Element 'e' has a horizontal force arrow pointing right and a stress value of 1998.11 MPa. The stress values are listed to the right of each element's square.	